VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANASANGAMA, BELAGAVI – 590018

An Internship Report on

Rooman Technologies

Submitted in partial fulfilment of the requirement for the award
Degree of Bachelor of Engineering
in
Computer Science & Engineering

Submitted by

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1AT21CS086

Internship carried out at

Internal Guide

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CERTIFICATE

This is to certified that this internship work presented by "**Rohith H Chavan"** bearing **USN "1AT21CS086" of Atria Institute of Technology, Bangalore** in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during 2024-25. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The internship report has been approved as it satisfies the academic requirements with respect to the internship report as prescribed for the said Degree.

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Issued on: 28/03/2025

This is to certify that Ms/Mr **Rohith H Chavan**, **CAN_34021174**, D/o, S/o **H L Halappa** has successfully completed the On-the-Job Training (OJT) for the course **AI – DevOPS Engineer** conducted by **Rooman Technologies facilitated** by **TC276326**, Visvesvaraya Technological University (TP_ID: TP199028) from 15/03/2025 to 28/03/2025 at Rooman Technologies Pvt. Ltd. Bengaluru.

Sincerely,

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DECLARATION

I, Rohith H Chavan (1AT21CS086) hereby declare that this internship work is carried out under the guidance of **Dr. Farhana Kausar** Associate Professor, Dept. of CS&E. This internship work is submitted to Visvesvaraya Technological University in partial fulfilment of the requirement for the award of degree of Bachelor of Engineering in Computer Science & Engineering for the academic year 2024-25.

Place: Bangalore

Date:

Signature of Student

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ABSTRACT OF INTERNSHIP

During my internship at Rooman Technologies, which commenced in September 2024, I gained hands-on experience in Artificial Intelligence (AI) and DevOps practices, working on real-time projects and structured learning modules designed to bridge academic knowledge with industry requirements.

The internship began with onboarding and foundational training in machine learning algorithms, Python programming, and DevOps tools such as Git, Docker, Jenkins, and Kubernetes. I actively participated in workshops and collaborative sessions focusing on CI/CD pipeline implementation, container orchestration, and version control management.

On the AI front, I engaged in developing and deploying ML models, data preprocessing, and performance evaluation using tools like Scikit-learn, Pandas, and TensorFlow. I was involved in building mini-projects that demonstrated the integration of AI models into production environments using DevOps workflows.

As part of the DevOps track, I contributed to setting up automated deployment pipelines, containerizing applications using Docker, and managing microservices using Kubernetes. I also documented scripts and deployment strategies for team reference and assisted in troubleshooting integration and deployment issues.

The internship emphasized both technical proficiency and collaborative problem-solving, allowing me to enhance my skills in AI model development, infrastructure automation, and software lifecycle management. This experience provided a solid foundation in building scalable and efficient AI-powered applications in a DevOps environment.

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Chapter 1

ABOUT THE COMPANY

Chapter 1

ABOUT THE COMPANY

Rooman Technologies is a leading IT training and services organization in India, specializing in cutting-edge technologies such as Artificial Intelligence (AI), DevOps, Cybersecurity, Cloud Computing, and Network Engineering. With a mission to empower professionals through industry- relevant skills and practical training, Rooman has played a pivotal role in shaping tech talent across the country.

Key Highlights

- **Purpose:** Rooman Technologies is committed to bridging the skill gap in emerging technologies by offering career-focused training and hands-on experience in AI, DevOps, and other high-demand IT domains.
- **Offerings:** The organization provides in-depth training programs in AI/ML model development, automation using DevOps tools, cloud platforms like AWS and Azure, and cybersecurity best practices. Training includes live projects, simulations, and certifications.
- **Presence in India:** Headquartered in Bengaluru, Rooman has a pan-India presence with over 200+ training centers and technology labs across urban and rural locations, helping students and professionals access advanced IT skills.
- **Industry Alignment:** Training programs are aligned with National Skill Development Corporation (NSDC) and Skill India initiatives, ensuring graduates meet the evolving needs of the IT industry.
- **Corporate Collaboration:** Rooman collaborates with organizations to offer workforce development solutions, including employee upskilling, campus-to-corporate transitions, and client-specific DevOps and AI project development support.
- **Student Impact:** Rooman has trained and placed over 1 million students, with a strong track record in preparing candidates for roles such as AI Developers, DevOps Engineers, Data Analysts, and Cloud Administrators.

Business Areas

- AI & Machine Learning: Focus on data preprocessing, supervised/unsupervised learning, model training, and deployment. Real-time projects using Python, Scikit-learn, TensorFlow, and NLP tools are core to the curriculum.

- DevOps & Automation: Offers hands-on training in CI/CD pipelines, Docker, Kubernetes, Git, Jenkins, and Ansible, enabling automation of software development and IT operations.

- Cloud Technologies: Teaches cloud deployment and monitoring using AWS, Azure, and Google Cloud, ensuring learners can build scalable and resilient cloud-native applications.

- Cybersecurity & Networking: Covers essential protocols, ethical hacking, firewalls, and secure infrastructure configuration with real-world labs and simulation tools.

Notable Projects

- Development and deployment of AI-powered chatbots for local businesses.
- Building end-to-end automated deployment pipelines for student and client projects using DevOps toolchains.
- Workshops and hackathons in collaboration with universities and tech companies to foster real-world problem-solving.

Sustainability Initiatives

- Digital Inclusion: Provides subsidized or free training to students from economically weaker sections, promoting digital literacy and job readiness.
- Skill Development Missions: Active participant in Skill India, PMKVY, and Digital India campaigns.
- Women in Tech: Focused programs to empower women through tech training in AI, DevOps, and Cloud Computing.

Chapter 2

ABOUT THE DEPARTMENT

Chapter 2

ABOUT THE DEPARTMENT

The AI & DevOps Department at Rooman Technologies plays a pivotal role in enabling innovation, automation, and operational efficiency across the organization. By integrating artificial intelligence with DevOps practices, the department ensures scalable, secure, and high-performance IT solutions.

Key Functions and Responsibilities

1. AI and Machine Learning Solutions:

- The department develops and deploys AI/ML models for predictive analytics, automation, and intelligent decision-making. Applications range from customer service automation to smart monitoring systems.

2. DevOps and CI/CD Implementation:

- The team is responsible for automating software development lifecycles using Continuous Integration and Continuous Deployment (CI/CD) pipelines. Tools like Jenkins, Git, Docker, and Kubernetes are employed to streamline and standardize development processes.

3. Cloud Infrastructure Management:

- Managing cloud platforms such as AWS, Azure, and Google Cloud is central to operations. This includes provisioning infrastructure as code (IaC), cost optimization, and high availability configurations.

4. Cybersecurity and Compliance:

- The department ensures secure deployment environments by enforcing security best practices, conducting audits, and complying with data protection regulations. This includes threat detection and vulnerability management.

5. Monitoring and Performance Optimization:

- Using tools like Prometheus, Grafana, and ELK stack, the department monitors applications and infrastructure to ensure optimal performance and quick resolution of issues.

Collaboration and Innovation

- **Interdepartmental Collaboration:**
  - The AI & DevOps team works closely with development, QA, and operations teams to foster a culture of collaboration and shared responsibility for product delivery.
- **Cutting-edge Innovation:**
  - The department leads research and implementation of emerging technologies including MLOps, serverless architecture, and AI-based system optimization to drive continuous innovation.

Impact on Business Operations

- **Operational Efficiency:**
  - Automated workflows and smart analytics lead to faster deployments, reduced downtime, and enhanced decision-making.
- **Risk Mitigation and Reliability:**
  - By integrating robust security practices and continuous monitoring, the department ensures business continuity and resilience.

Conclusion

The **AI & DevOps Department at Rooman Technologies** is a cornerstone of the company's digital strategy, enabling scalable solutions and continuous innovation. Through technical expertise and a commitment to excellence, the team ensures that Rooman Technologies remains agile, secure, and ahead in the competitive tech landscape.

Chapter 3

TASK PERFORMED

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TASK PERFORMED

During my internship at **Rooman Technologies**, I gained extensive hands-on experience in areas related to DevOps practices, automation, data analytics, and system performance optimization. Here are some of the key tasks I performed:

Key Tasks and Responsibilities

• Understanding AI and DevOps Workflows:

- Gained knowledge in implementing AI and automation solutions in the software development lifecycle, from code integration to continuous deployment.

• Knowledge Transfer Sessions on DevOps Tools:

- Participated in knowledge transfer (KT) sessions covering critical DevOps tools and methodologies, including Jenkins, Docker, Kubernetes, and GitLab, which are vital for CI/CD processes.

• Hands-On Tasks with DevOps Systems and Platforms:

- Worked with key DevOps systems like Jenkins and Kubernetes, where I helped automate the deployment process, monitored system health, and ensured smooth integration across development and operations.

• Incident Creation and Management in DevOps Tools:

- Gained experience in managing incidents and troubleshooting application issues using DevOps monitoring tools. This included tracking failures, debugging, and implementing fixes in real-time.

• Data Analysis and Reporting with PowerBI:

- Conducted data analysis using PowerBI, focusing on incident management and system performance metrics. Generated reports to help optimize DevOps operations and identify areas for improvement.

• Documentation for Automation and Change Management:

- Assisted in preparing documentation for change management, specifically focusing on automation workflows and incident handling in a cloud-native environment.

• Workflow Planning for DevOps Automation Workshops:

- Developed workflows and plans for upcoming DevOps automation workshops, where regional teams from various markets, including North America and Europe, collaborated to share best practices.

• Participation in DevOps Automation Workshops:

- Actively participated in workshops on DevOps automation, collaborating with teams from different regions to ensure streamlined, effective implementation of automation tools and processes.

• Follow-up Discussions on Automation Best Practices:

- Engaged in follow-up discussions on AI-driven automation and best practices with cross- functional teams to enhance efficiency and ensure the integration of cutting-edge technologies into the DevOps pipeline.

• User Management and Compliance with Governance Systems:

- Worked on organizing and managing user roles in governance systems, ensuring proper compliance and alignment with the organization's IT policies.

• Performance Analysis of DevOps and AI Systems:

- Analyzed incidents and performance data related to DevOps and AI systems, identifying trends in system performance, optimizing message processing, and improving operator efficiency.

Chapter 4

REFLECTION NOTES

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4.1 Internship Experience

My internship at Rooman Technologies was an eye-opening and transformative experience, giving me the opportunity to work in the fast-paced world of AI and DevOps. The environment encouraged innovation and growth, providing me with a blend of hands-on technical experience and insight into real-world challenges. Through the guidance of professionals, I applied theoretical knowledge to practical scenarios. The tasks and projects I was involved in, from DevOps automation to AI integrations, expanded my technical capabilities and gave me a deeper understanding of cloud computing, CI/CD pipelines, and machine learning. This internship served as a pivotal milestone in my career, sharpening my technical and interpersonal skills.

4.2 Technical Outcomes

Throughout the internship, I developed a strong foundation in several key technical areas:

- **AI & Machine Learning Integration:** I gained exposure to integrating AI models with DevOps pipelines, including automation of model deployments and monitoring the performance of AI applications in production environments.
- **DevOps & Automation:** I was introduced to the full DevOps lifecycle, from code integration to continuous deployment using tools like Jenkins, Docker, Kubernetes, and GitLab. I learned how automation accelerates the software delivery process and ensures system stability.
- **Data Analysis & Reporting:** Using PowerBI, I performed data analysis to track DevOps pipeline performance and incident trends, generating actionable insights that contributed to improving system efficiency and incident resolution strategies.
- **Documentation & Workflow Development:** I was involved in documenting change management processes, creating workflows for automation workshops, and refining my technical writing skills, which contributed to smoother project management and execution.

4.3 Non-Technical Outcomes

The internship also contributed to my personal and professional growth in several non-technical areas:

4.3.1 Communication Skills

Effective communication was vital throughout my internship. I worked on improving my ability to present complex technical concepts to non-technical stakeholders and fine-tuned my communication skills for diverse audiences. Through team discussions, documentation creation, and presenting findings, I grew more confident in my ability to articulate ideas clearly and concisely.

4.3.2 Adapatability Skills

In the dynamic world of AI and DevOps, adaptability was key. The fast-changing environment required me to quickly adjust to new tools, technologies, and workflows. Whether learning new automation tools or troubleshooting issues in real-time, I developed the ability to handle multiple tasks effectively, prioritize demands, and adapt to evolving project needs.

4.3.3 Work Ethics

Rooman Technologies provided a professional environment that fostered a strong sense of work ethics. I learned the importance of punctuality, responsibility, and maintaining high standards of integrity. The experience helped me understand how critical professionalism is in ensuring the success of projects and in building trust with colleagues and clients.

4.3.4 Responsibility

Being entrusted with critical tasks, such as automation process implementations and incident management, taught me accountability and ownership. I learned to proactively tackle challenges and manage responsibilities, ensuring that deadlines were met and results were delivered to high standards. This experience helped me grow both as a contributor and as a leader in the team.

4.3.5 Social Connect Skills

The collaborative nature of the work, participation in workshops, and follow-up discussions not only improved my teamwork but also strengthened my ability to empathize, communicate effectively, and work toward common goals.

REFERENCES

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- https://rooman.com/
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